

CostVision PCB Image -> BOM Should-Cost

Automotive 77GHz Radar Module (NXP S32R294) — COLD RE-RUN (Run 2)

China (Shenzhen / Suzhou) · 150,000/yr · 5-year program · vision pipeline (Haiku classify + ASIL -> Sonnet BOM)

IMAGE -> BOM (AI)

GBP 72.28

per board, China 150k

ENGINEERING (corrected)

GBP 66.69

8% vs AI

SAFETY

ASIL-C

ISO 26262 ASIL

1. Board specification — AI vision vs measured ground truth

Parameter	AI (from photos)	Ground truth (board panel)	Verdict
Layers	8	8 layer	OK
Dimensions	85 x 155 mm (131.8 cm2)	87.8 x 48.9 mm (42.9 cm2)	AI ~3.1x area
Surface finish	ENIG	Immersion Silver	AI over-spec
Technology	HDI RIGID	Rigid FR4 (1.31 mm)	~
Weight	-	26.4 g	ref

Board size has no scale reference in a photo, so absolute dimensions are the AI's weakest estimate. Because fab is a small fraction of a component-heavy automotive board, this only moves the total ~8% (fab GBP 6.07 -> GBP 1.98 at the true 42.9 cm2). Layer count, BOM, ASIL and program pricing are unaffected.

2. Should-cost breakdown (per board, China @ 150k)

Cost element	Image -> BOM (AI)	Engineering (corrected)	Notes
PCB fabrication	GBP 6.07	GBP 1.98	8-layer, area-driven
SMT assembly	GBP 5.67	GBP 5.67	274 placements, 1 BGA
BOM (components)	GBP 56.40	GBP 56.40	AEC-Q automotive grade
Logistics	GBP 2.64	GBP 2.64	ex-China
TOTAL / board	GBP 72.28	GBP 66.69	8%

3. Bill of Materials (AI-extracted, 18 lines · 13 need verification)

RefDes	Description	Qty	Unit GBP	Ext GBP
U1	NXP S32R294 automotive radar MCU (FS32R294KCMJ	1	18.000	18.00
U2	77GHz FMCW radar transceiver MMIC (under glob-	1	4.000	4.00
U3	Automotive system basis chip / PMIC (FS65/FS85	1	4.840	4.84
U4	Automotive safety PMIC (TLF35584-class)	1	4.524	4.52
U5, U6	CAN FD transceiver (TJA1044-class)	2	0.827	1.65
U7-U9	Automotive LDO voltage regulator	3	3.080	9.24
U10	Unmarked support IC (LVDS/gatekeeper/interface	1	4.000	4.00
U11	Secondary interface IC (bottom side, near ref	1	1.760	1.76
Y1	Automotive TCXO oscillator (SMD can)	1	2.000	2.00
C1, C2	Aluminium electrolytic capacitor 100V (JW 100u	2	1.936	3.87
D1-D6	Automotive TVS/protection diode	6	0.264	1.58
Q1-Q4	Automotive MOSFET/small-signal transistor	4	0.440	1.76
R1-R90	AEC-Q200 automotive resistor	90	0.005	0.48
BOM total				GBP 64.10

RefDes	Description	Qty	Unit GBP	Ext GBP
C10-C119	AEC-Q200 X7R MLCC capacitor	110	0.013	1.45
C120-C159	AEC-Q200 X7R MLCC capacitor (power rail decoup	40	0.026	1.06
L1-L12	AEC-Q200 automotive inductor/ferrite bead	12	0.030	0.36
J1, J2	2x9 THT pin header (board-to-board interconnec	2	1.320	2.64
J3	Small 2x5 THT header (bottom side, ref 0338)	1	0.880	0.88
BOM total				GBP 64.10

4. Functional safety (ISO 26262) & automotive NRE

ASIL: ASIL-C — NXP S32R automotive radar MCU with integrated safety features, dual large electrolytic capacitors for redundant power conditioning, dense mixed-signal IC population for sensor signal processing, organized RF signal routing with EMI shielding, and automotive-grade component layout indicate a safety-critical radar perception system. The presence of multiple power distribution circuits, thermal management design, and high-frequency signal conditioning circuits suggest ASIL-C level functional safety requirements typical of advanced driver assistance systems (ADAS) radar processing.

Safety functions: Radar signal acquisition and processing; Object detection and tracking; Collision avoidance support; Power supply monitoring and fault detection; Thermal management and protection

Automotive NRE (one-time, amortised separately): PPAP GBP 11,200 · FMEA GBP 20,200 · DVP&R GBP 31,400 · ASIL audit GBP 11,200 = GBP 74,000 total.

5. Program pricing (5-yr) & regional options

Program contract pricing at this volume: spot BOM GBP 64.09 -> program BOM GBP 32.04 (50% via tier1 contract).

Region	Fab	Assembly	BOM	Total/board
China	GBP 6.07	GBP 5.67	GBP 56.40	GBP 72.28
Vietnam	GBP 7.67	GBP 4.92	GBP 58.96	GBP 77.48
Malaysia	GBP 9.01	GBP 7.75	GBP 58.32	GBP 80.72
India	GBP 9.01	GBP 6.43	GBP 60.89	GBP 82.84
Thailand	GBP 9.77	GBP 7.34	GBP 59.60	GBP 82.60
Taiwan	GBP 15.49	GBP 11.54	GBP 57.68	GBP 89.00
Mexico	GBP 13.95	GBP 8.88	GBP 62.81	GBP 90.51
South Korea	GBP 18.75	GBP 13.23	GBP 58.96	GBP 92.05

6. Accuracy & QA

- Result GBP 72.28/board is defensible for an ASIL-C 77GHz automotive radar module at 150k (component-heavy: BOM is 78% of cost). Confidence band total GBP 60.54-GBP 77.33-GBP 114.67.
- Board DIMENSIONS are AI-estimated with no scale reference and read ~3.1x the true area; confirm against the fab drawing. Corrected here to 42.9 cm2 (fab GBP 1.98).
- Surface finish read as ENIG; the board panel says immersion silver (slightly cheaper). Transceiver MMIC is glob-top (die hidden) so its price is an estimate.
- 13 of 18 BOM lines are vision-estimated (not catalogue-confirmed) — expected for image-only costing; confirm high-value ICs (radar MCU, transceiver) against a distributor before quoting.

7. Bugs fixed in this run

This board initially mis-classified as non-automotive, which silently blanked ASIL, program pricing and AEC-Q grading. Fixed: (1) domain classifier now recognises automotive radar (S32R MCU + antenna array) and promotes deterministically from the IC marking; (2) ASIL parser salvages the level instead of blanking to Unknown on a JSON hiccup (now ASIL-C); (3) program pricing now applies the 50% tier-1 contract multiplier for the 5-yr program. All verified live.